Exploratory Analysis of Academic, Behavioral, and Social Factors Associated with Student Performance

Problem Definition & Motivation

Many times teachers are quick to be blamed for student performance in schools. In a recent article by the Honolulu Star-Advertiser, it was reported that only a small percentage of Hawaii students were meeting proficiency standards in math. The comments section was quick to blame teachers for the shortcomings of the students. However, many teachers work extremely hard and put in more than the salaried 40-hour work week in order to help their students as much as possible, despite not always knowing what goes on beyond the classroom.

This report aims to determine whether there are factors outside of the classroom that are associated with student performance. In addition, this project explores whether student performance can be predicted using academic, behavioral, and social variables so that classrooms can become more data-driven and teachers can better understand the students they work with.

Research Questions

1. How strongly do prior academic grades predict a student's final performance?
2. Does increased study time improve student performance?
3. How do alcohol consumption and social activity affect student performance? Is there a negative relationship as commonly assumed?
4. What role do family background and educational support play in a student's academic performance?
5. Which variables demonstrate the strongest overall associations with final academic performance?

Dataset

The dataset used for this project comes from the UCI Machine Learning Repository. The dataset contains student grades along with demographic, social, behavioral, and school-related features. The attributes were collected using school reports and student questionnaires. This analysis specifically focuses on student performance in mathematics.

The target variable for this dataset is G3, which represents the student's final grade for the year. Other important variables include G1 and G2, which represent grades from the first and second grading periods. The dataset also includes categorical features such as school support, family support, internet access, and higher education aspirations. Numerical features such as study time, free time, alcohol consumption, and absences are also included.

This dataset provides a strong starting point for exploring the research questions because it contains a wide range of variables that may influence student performance both inside and outside of the classroom.

There was no major cleaning required for this dataset because there were no missing values or duplicate rows. During preprocessing, I created additional engineered features to simplify analysis. This included combining weekday and weekend alcohol consumption into a single total_alcohol feature and calculating an average parent education level using the mother's and father's education variables. These changes helped standardize the analysis and made it easier to compare broader trends across the dataset.

Methodology

To determine which attributes were most strongly related to student academic performance, I first performed exploratory data analysis (EDA) on the dataset. The analysis was divided into three major groups: academic factors, behavioral and lifestyle factors, and family and social factors. Breaking the analysis into these categories made it easier to identify which groups of variables had the strongest relationships with student outcomes.

The analysis included histograms, boxplots, regression plots, grouped statistical summaries, and correlation heatmaps. These visualizations helped identify patterns and trends between variables and the final grade (G3).

After exploratory analysis, I used both simple linear regression and multiple linear regression to quantify relationships between selected variables and final student performance. Simple linear regression was used to measure how strongly second-period grades (G2) predicted final grades (G3). Multiple linear regression was then used to evaluate how academic and behavioral variables interacted when analyzed together.

Experiments & Results

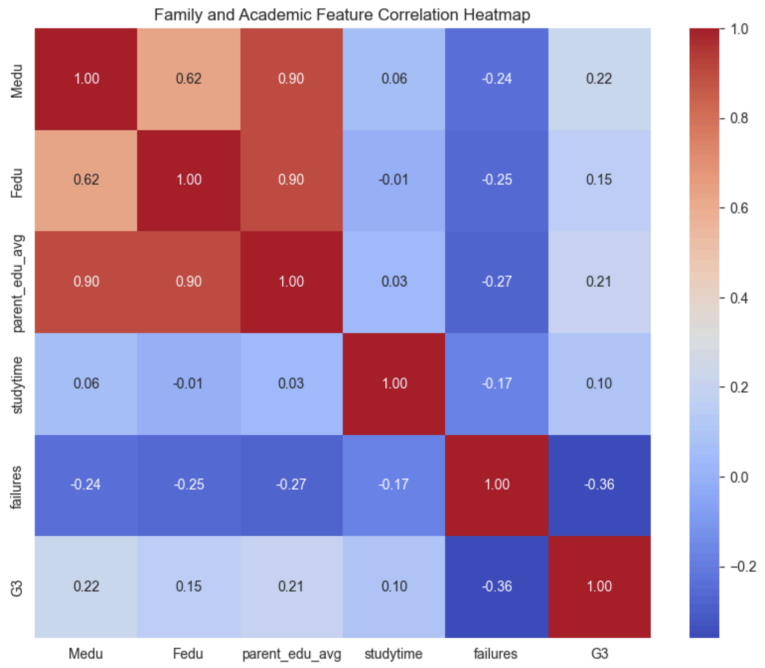
Academic Factors

The first major insight from the analysis was that academic variables played the strongest role in determining student performance. G1 and G2 were by far the strongest indicators when predicting the final G3 score. The regression plots and correlation analysis showed extremely strong positive relationships between earlier grades and final academic outcomes, suggesting that student performance remains relatively consistent across grading periods.

Past academic failures also showed a strong negative relationship with final grades. Students with a higher number of previous failures generally performed substantially worse academically. Study time

demonstrated a moderate positive relationship with final performance, although the relationship was noticeably weaker than the relationship observed with prior grades.

Absences showed a weaker negative relationship with final grades. While students with very high absence counts often performed worse academically, there was still significant variability between students, suggesting that attendance alone does not fully explain student outcomes.

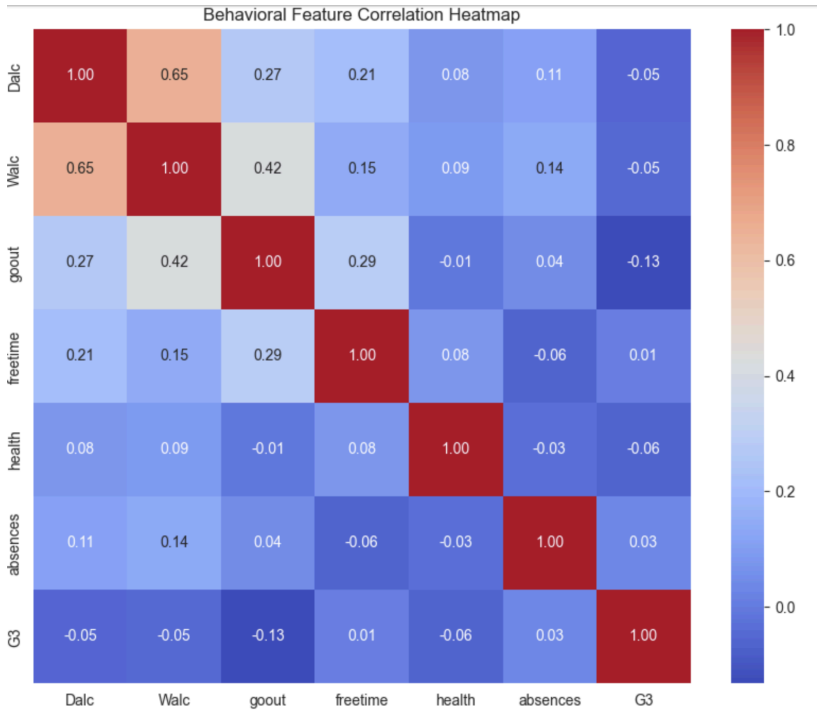


Behavioral & Lifestyle Factors

Behavioral and lifestyle variables demonstrated weaker relationships with academic performance compared to academic variables. Alcohol consumption and social activity both showed slight negative relationships with final grades, although the effects were much smaller than many people might assume.

Students with higher levels of alcohol consumption, especially when combining weekday and weekend drinking, tended to have somewhat lower academic performance on average. Similarly, students who reported going out with friends more frequently generally showed slightly lower grades. However, there was still heavy overlap between groups, indicating that these variables alone are not strong predictors of student performance.

One important finding from this section was that many public assumptions about student behavior appear to oversimplify academic performance. While behavioral choices may contribute to outcomes, they do not independently determine success or failure in school.

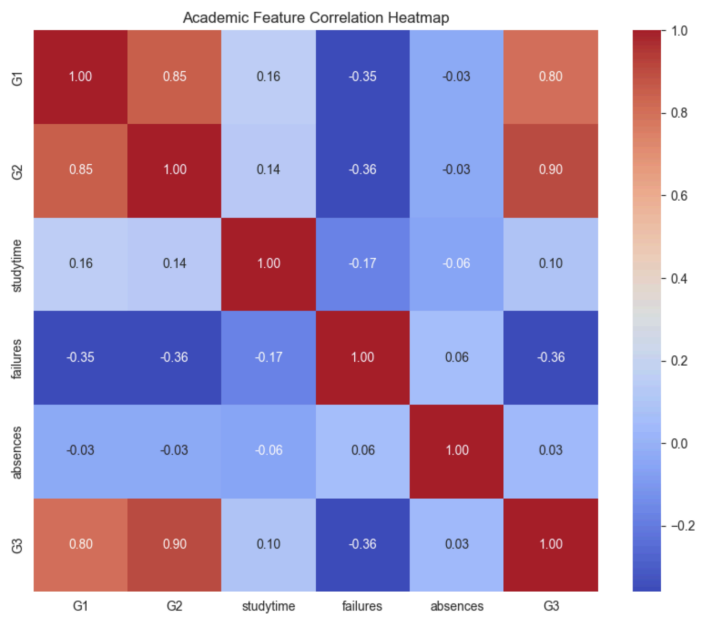


Family & Social Factors

Family and social support variables demonstrated moderate relationships with academic performance. Students with parents who had higher education levels generally achieved slightly higher final grades on average. Students who expressed a desire to pursue higher education also tended to perform substantially better academically.

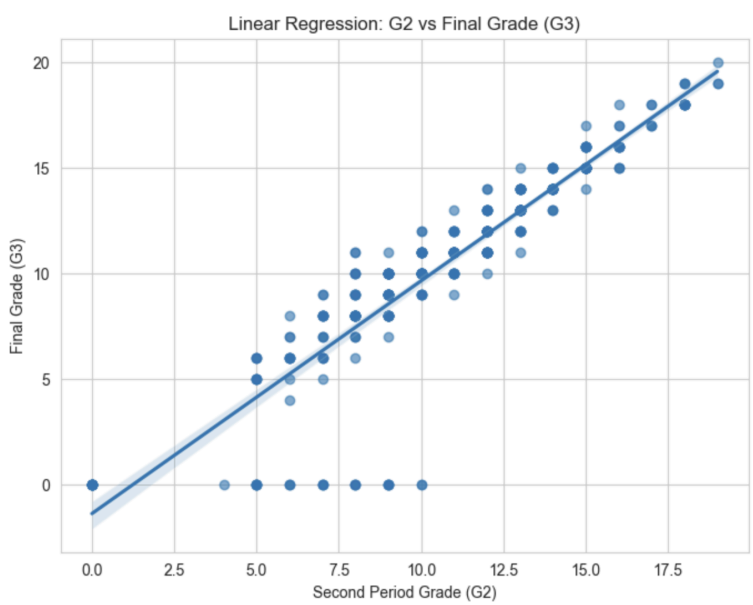
An especially interesting finding involved school educational support. Students receiving additional school support often had lower academic outcomes. However, this likely reflects the fact that struggling students are more likely to receive additional education services rather than suggesting that support itself negatively affects performance.

Family support and family relationship quality showed weaker but still observable positive relationships with academic outcomes. These findings suggest that a student's broader environment and support systems may contribute to performance, although academic history still remains the strongest overall predictor.



Linear Regression Analysis

Simple linear regression demonstrated a very strong relationship between second-period grades (G2) and final grades (G3). The regression model produced a high R^2 value, indicating that a large proportion of the variation in final grades could be explained by second-period performance alone.



Multiple linear regression further reinforced the earlier exploratory analysis findings. When academic and behavioral variables were analyzed simultaneously, prior academic grades (G1 and G2)

remained the strongest predictors of final performance. Behavioral variables such as alcohol consumption and social activity became much weaker predictors within the multivariate model.

This suggests that although behavioral and lifestyle factors may influence academic outcomes to some extent, prior academic achievement remains the most consistent and dominant indicator of future student success within this dataset.

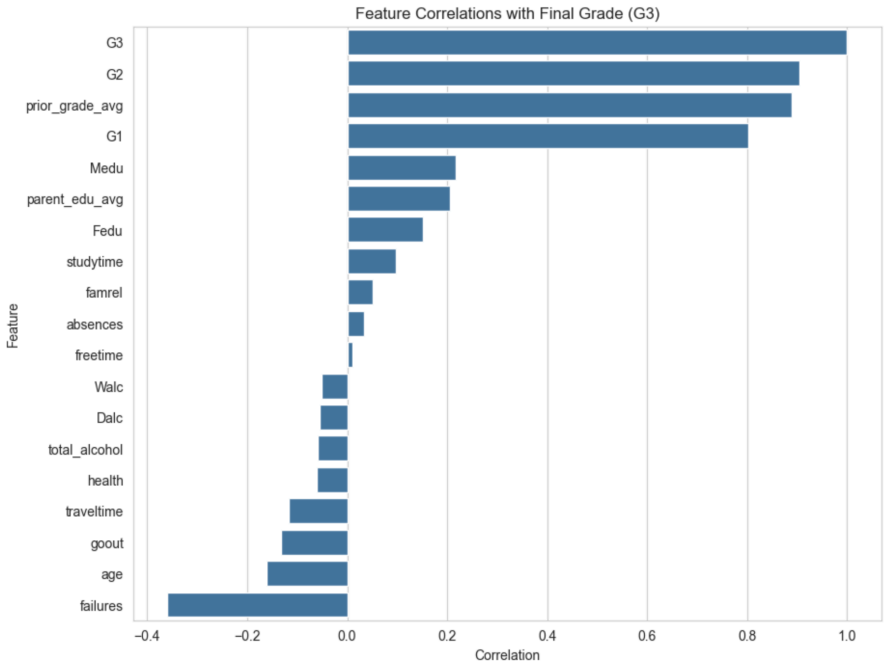
Discussion & Insights

Overall, the analysis demonstrated that student academic performance is influenced by a combination of academic, behavioral, and social factors. However, prior academic achievement consistently showed the strongest relationship with final student outcomes.

One of the most important takeaways from this project is that student performance cannot simply be reduced to a single factor such as teacher quality, alcohol use, or social activity. The results suggest that academic performance is multifactorial and shaped by a broader combination of educational history, support systems, lifestyle behaviors, and environmental influences.

There are several limitations to this analysis. First, the dataset is observational and cannot establish causal relationships between variables. This was apparent when looking at the educational school support relationship. Also, many behavioral variables were self-reported by students and may contain reporting bias or inaccuracies.

Future work could expand this project by incorporating additional socioeconomic variables, comparing multiple subject datasets, or monitoring student data to better understand how academic performance evolves over time.



AI Usage

I utilized ChatGPT when writing this report to clean up a lot of my writing. All of the analysis done in the Jupyter Notebook is mine and I wrote a base draft of the report and had ChatGPT clean it up for me. I also, however, consulted with ChatGPT when looking at how to lay out my analysis. I asked for recommendations regarding the best visualization methods for each feature and how to best convey the analysis.